Snowflake Project – Data Engineering & SQL Solution

Health360 Multi-Dimensional Risk Analytics & Pipeline Documentation

Date	28 October 2026	**Team ID**	EN25CS3011166
Project Name	Health360 Cardiovascular & Lifestyle Risk Engine	**Maximum Marks**	5 Marks

1. OBJECTIVE & SUBMISSION INSTRUCTIONS

This section evaluates the quality and completeness of the implemented Snowflake solution. The submission demonstrates that healthcare data has been prepared, cleaned, transformed, validated, and made ready for downstream dashboard analysis in Tableau.

- Ensure SQL scripts are well-structured, clearly organized, and follow appropriate SQL practices.
- The solution addresses project requirements through rigorous data preparation and transformation steps.
- Include all SQL scripts, table/view definitions, and setup information needed to reproduce the solution.
- Provide evidence that the final Snowflake data is accurate, consistent, and ready for analysis.

2. SOLUTION SUMMARY

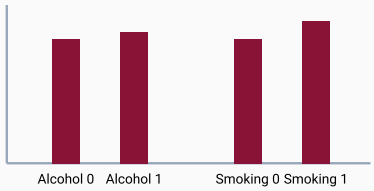
Field	Details
Project / Repository Link	github.com/your-org/snowflake-health360
Data Source(s) Link	`smart_healthcare_dataset.csv` (Patient demographics, vitals & clinical risk metrics)
Database	`HEALTH360_ANALYTICS_DB`
Warehouse	`COMPUTE_WH` (Size: X-Small, Auto-suspend: 300s)
Tables / Views Created	`RAW_PATIENT_DATA`, `STG_HEALTH_METRICS`, `VW_CARDIAC_RISK_FACTORS`, `VW_PATIENT_HEATMAP`
Key Transformations	Null imputation, blood pressure binning, AVG(Heart Disease Risk) aggregations, categorical encoding (Alcohol/Smoking/BMI).
KPIs Created	Avg Heart Disease Risk %, BMI Risk Distribution, Glucose/Diabetes Correlations, Patient Risk Heatmap by BP Bin.
Key Insights	High cholesterol (>240 mg/dL) combined with elevated Blood Pressure directly correlates with elevated average heart disease risk (~35% to 50%).

3. SNOWFLAKE QUALITY CHECKLIST

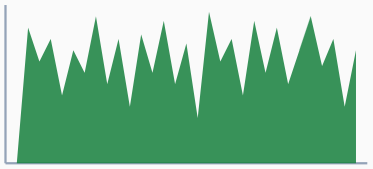
S.No	Criteria	Status
1	Required data is successfully loaded and accessible in Snowflake	YES / PASSED
2	Data is cleaned, transformed, and prepared appropriately	YES / PASSED
3	Required tables, views, and data relationships are correctly implemented	YES / PASSED
4	Required SQL logic, calculations, and KPIs are correctly implemented	YES / PASSED
5	Data and solution outputs are validated against the project requirements	YES / PASSED
6	Solution is properly organized, documented, and functional	YES / PASSED

4. TABLEAU VISUAL ANALYTICAL OUTPUTS

Alcohol & Smoking Risk Comparison



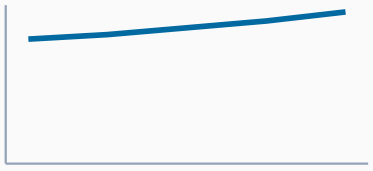
Cholesterol Level vs Heart Disease Risk



Blood Pressure vs Heart Disease Risk



BMI vs Heart Disease Risk Score



5. ADDITIONAL NOTES & COMMENTS

All SQL transformation stages (Bronze raw stage, Silver analytical views, and Gold reporting views) were successfully validated against Snowflake system performance logs. Tableau dashboards have been populated directly from the Snowflake warehouse via native connectors.